Poultry Feed Plant Intelligence System
Case Study

This document presents a comprehensive case study of the Poultry Feed Plant Intelligence System. The system integrates Industrial HMI, AWS Cloud Infrastructure, Backend Processing, and a Web-based Frontend Dashboard to monitor, control, and manage poultry feed production operations in real time.

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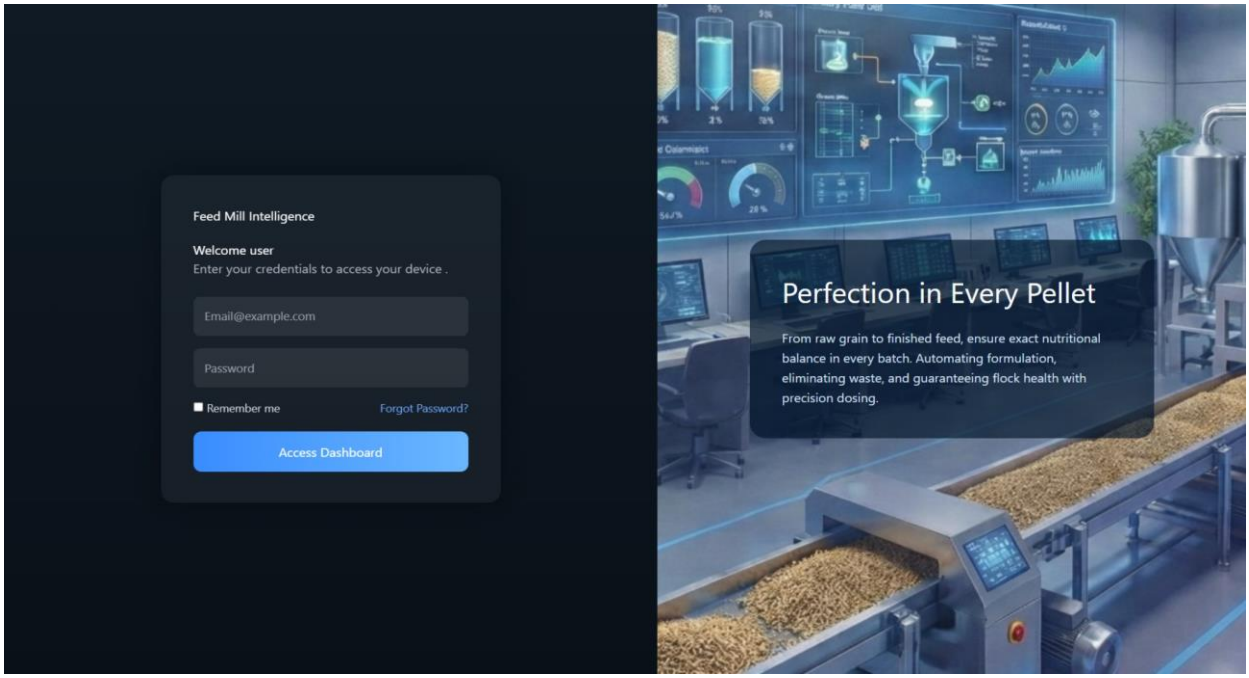
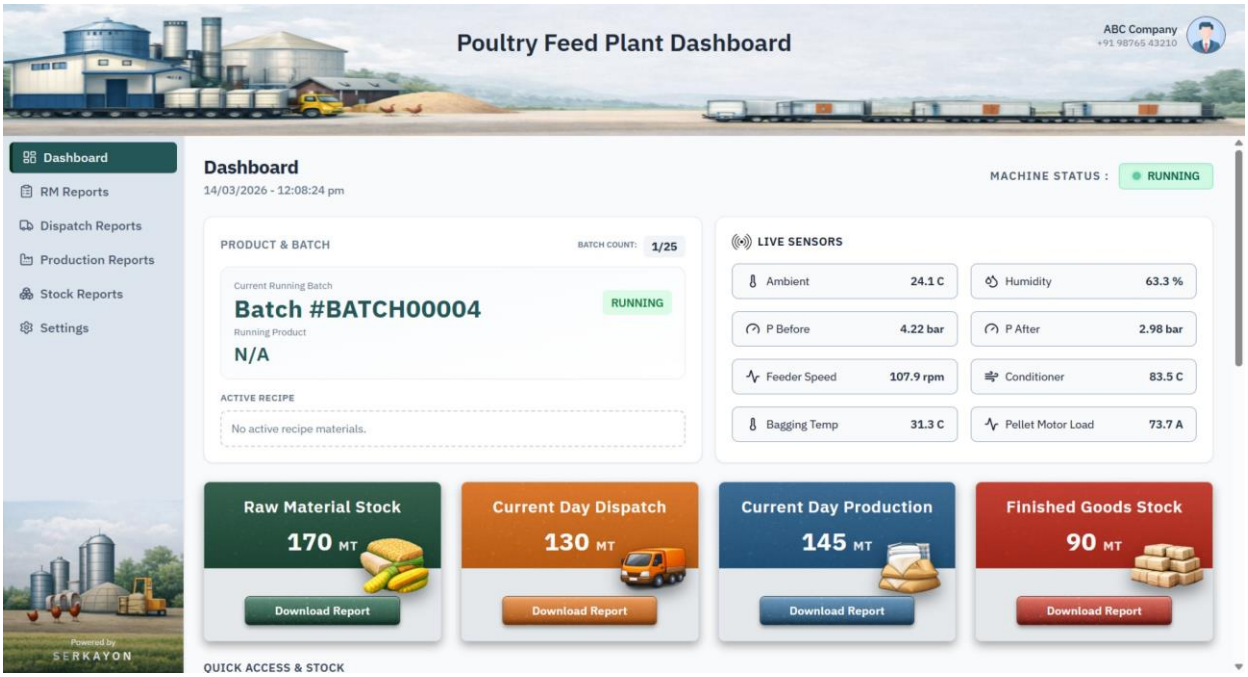
1. Login System

- The login interface provides secure authentication for authorized users to access the feed mill intelligence dashboard. Users enter credentials to access the system which connects to backend services hosted on AWS infrastructure.
- Security
- Key Features:
    - Credential authentication
    - Session management
    - Controlled system access
    - Secure device connectivity



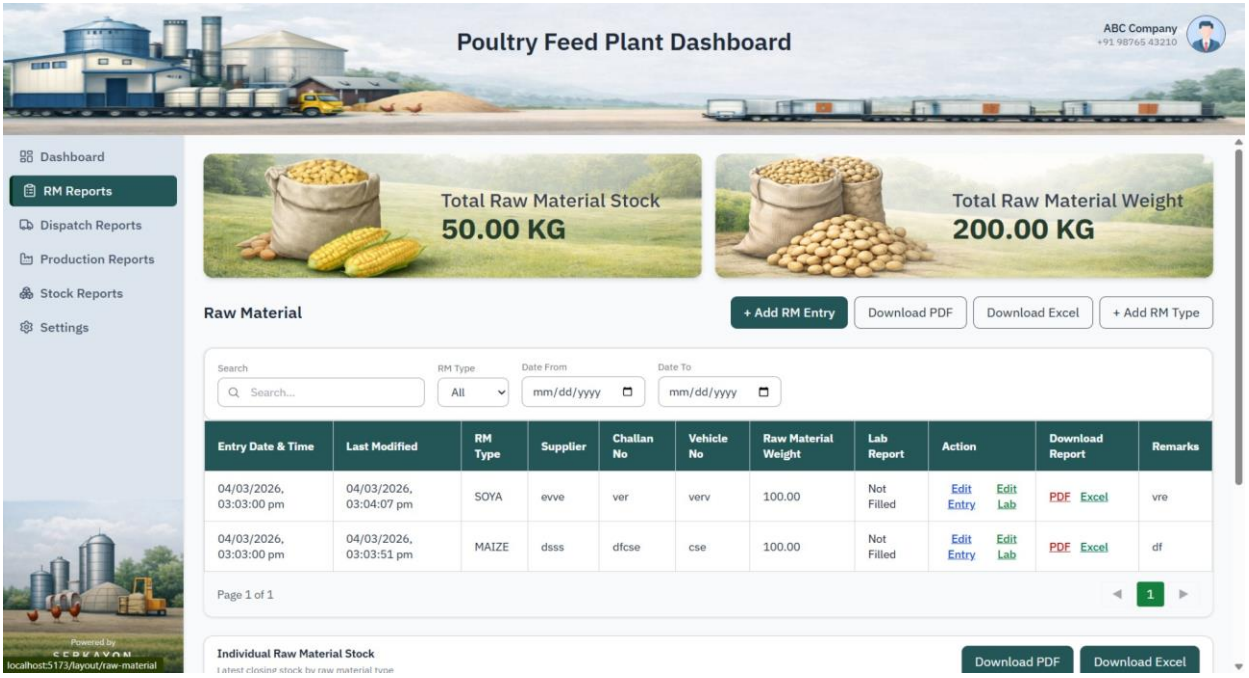
2. Dashboard Overview

- The central dashboard provides real time monitoring of the plant operations.
- Key Features:
    - Live machine status monitoring
    - Active batch visibility
    - Active recipe display
    - Sensor data monitoring
    - Production metrics overview
    - Downloadable daily reports with graphs



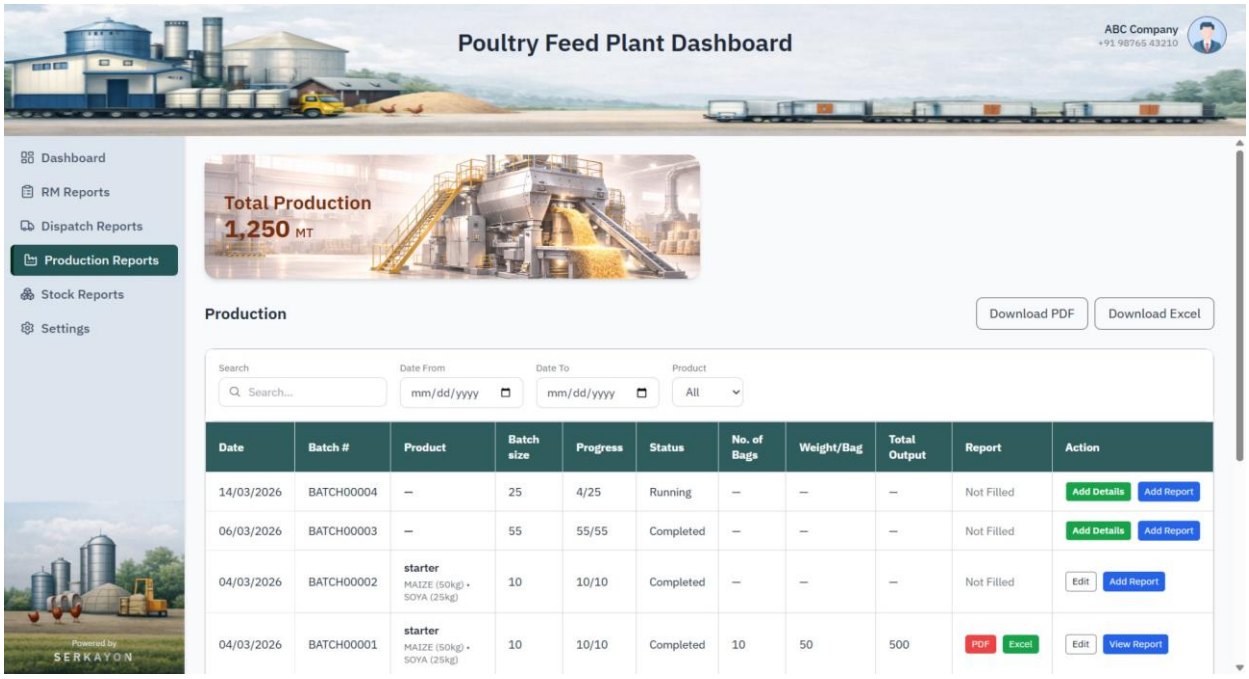
3. Raw Material Management

- Raw material management ensures proper tracking of incoming materials such as maize, soya, DDGS and other ingredients used in feed production.
- Key Features:
  - • Add raw material entries
  - • Record supplier and challan details
  - • Attach laboratory test reports
  - • Download RM reports in PDF and Excel
  - • Maintain weight tracking Accurate raw material stock is essential for recipe preparation and production planning.



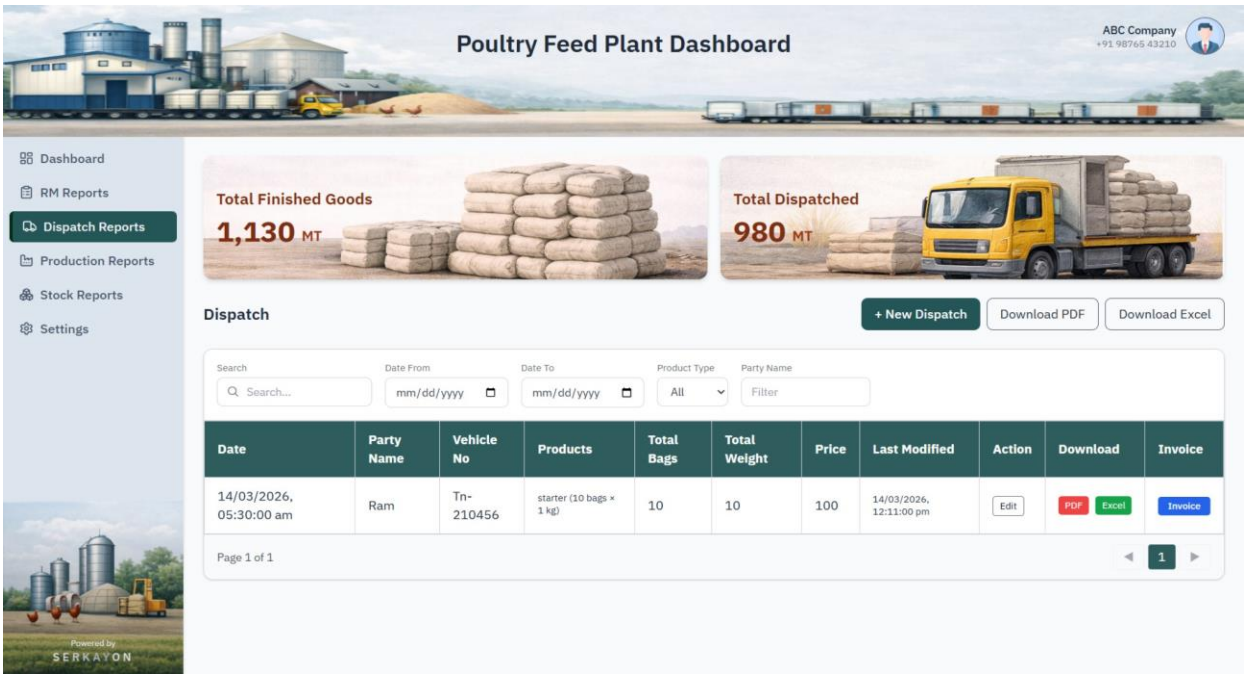
4. Production Management

- Production is executed based on predefined recipes using available raw materials.
- Key production functions include:
  - Batch tracking
  - Production progress monitoring
  - Output quantity calculation
  - Bag count and weight monitoring
  - Report generation Production data is synchronized with HMI batch execution to ensure consistency between the industrial machine and the cloud monitoring system.



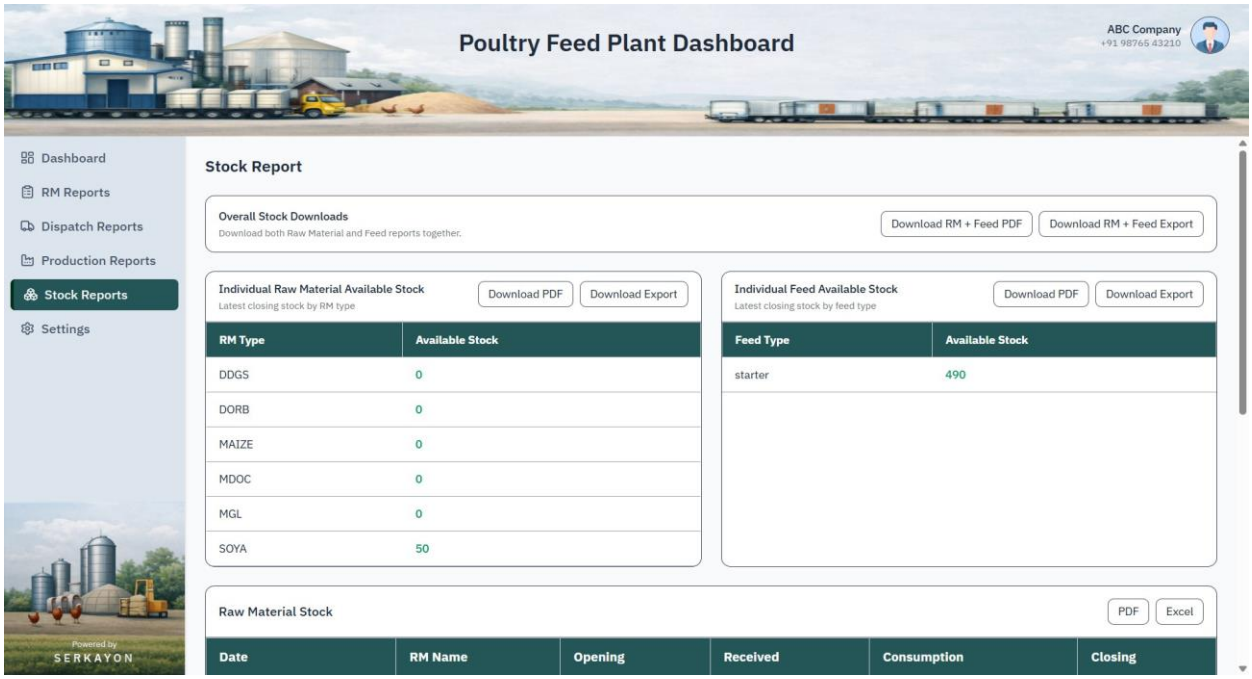
5. Dispatch Management

- Dispatch management handles distribution of finished goods produced in the plant.
- Functions:
  - • Dispatch entry creation
  - • Party and vehicle tracking
  - • Product quantity management
  - • Automatic invoice generation
  - • Exportable dispatch reports Only products produced in the production stage are available for dispatch.



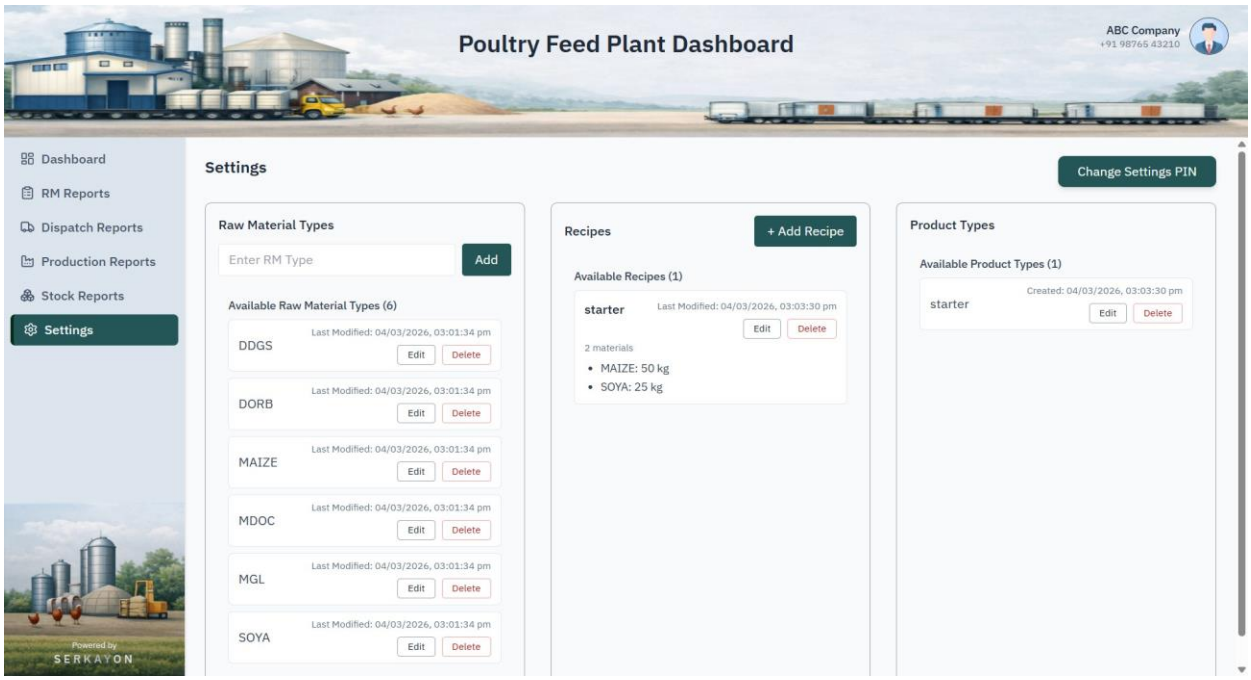
6. Stock Monitoring

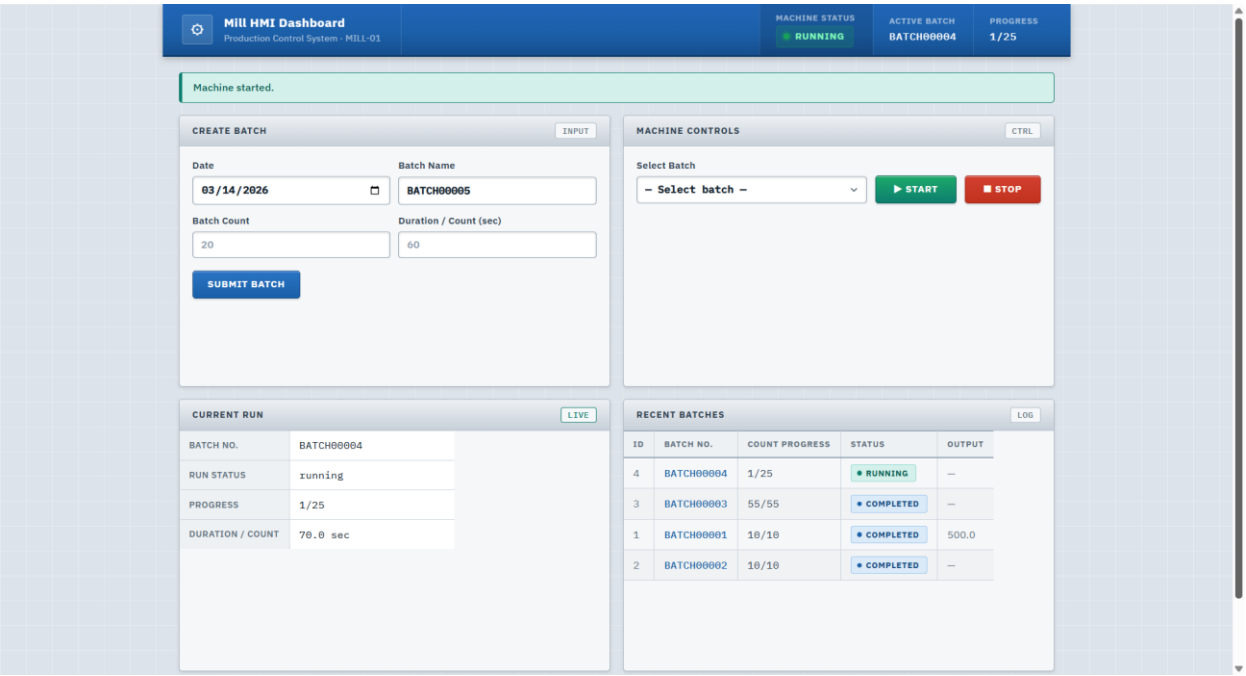
- Stock monitoring provides complete visibility into raw material and finished feed inventory. Features:
  - • Opening stock tracking
  - • Closing stock calculation
  - • Available stock display
  - • Individual and total stock reports
  - • Combined RM and Feed export reports Stock reports can be exported in Excel or PDF format for auditing and operational analysis.



7. System Settings

- System settings allow administrators to configure core operational data including:
  - Raw material types
  - Product types
  - Feed recipes
  - Batch formulation Recipes are critical because production batches rely on the defined material composition.



8. HMI Integration Architecture

- The industrial HMI (Human Machine Interface) communicates directly with the cloud platform.
- Workflow:
    - Operator creates a batch in the HMI system.
    - Batch ID, batch name, and recipe details are transmitted to AWS cloud services.
    - Backend services process and store batch data.
    - The frontend dashboard retrieves the processed data.
    - Live updates display machine status, active batch, and production progress.
    - This architecture ensures real time monitoring and centralized control of plant operations.